

two_chunk_planner CLI reference (en)

two_chunk_planner plan CLI reference

Version 0.2.0 — GPL-3.0-or-later. This reference describes the standalone canonical 90°×90° AB-central/AC-supported-left planner. It has no project-root, manifest, SPECfEM source, or patch-verification mode. The parser inventory is generated in [generated/cli_options.json](#) and is checked against this page.

Running modes

Standalone: supply source, stations, and phases or a time endpoint. The bundled profile supplies planning defaults only. **Standalone plus Par_file:** add `--par-file /any/path/Par_file`; only resource-relevant values are read. Neither mode verifies a user's SPECfEM source or accepted patch. Use the separate patch package before production simulations.

```
two-chunk-planner plan --source 0 0 50 --stations-csv stations.csv \
--analysis-window 0 1900 --output plan_out
```

Every option has no short form. `--output` is required; all other defaults below are parser defaults. Exactly one source form and exactly one station form are required. At least one of phases, analysis window, or target-energy end is required. An existing output directory is always refused.

The installed `two-chunk-planner` entry point and `PYTHONPATH=packages/two_chunk_planner/src /usr/bin/python3 -m two_chunk_planner` use this same parser. The latter is a repository-development invocation, not a separate CLI mode.

1. Source and station input

--cmtsolution

Short form: none. Type: path. Optional but mutually exclusive with `--source`; one is required. Reads SPECfEM CMTSOLUTION (PDE header plus labelled records). Example: `--cmtsolution DATA/CMTSOLUTION`. Invalid/missing fields or a zero moment tensor stop planning. It supplies source coordinates and provenance.

--source

Short form: none. Type: three floats LAT LON DEPTH KM; no default. Mutually exclusive with `--cmtsolution`; one is required. Example: `--source 0 -125 50`. Latitude/depth must be finite and depth non-negative. It bypasses CMTSOLUTION parsing only; it does not create an input file.

--stations

Short form: none. Type: path; mutually exclusive with `--stations-csv`; one is required. Example: `--stations DATA/STATIONS`. Reads six whitespace fields per receiver. Duplicate network/station IDs or malformed numeric coordinates fail.

--stations-csv

Short form: none. Type: path; mutually exclusive with `--stations`; one is required. Example: `--stations-csv stations.csv`. Required columns are `network,station,latitude_deg,longitude_deg`; `elevation/burial` are optional.

2. Phase, analysis window, and target end

--phases

Short form: none. Type: comma-separated names; default none. Used only by phase-aware; example `--phases Pdiff,Sdiff`. It requests exactly these TauP phases—no substitute phase is used. It may be omitted only when a window/end is provided.

--analysis-window

Short form: none. Type: two floats START_S END_S; default none. Example: `--analysis-window 0 1900`. Its end becomes the advisory boundary-time comparison endpoint unless `--target-energy-end-s` is supplied.

--target-energy-end-s

Short form: none. Type: float seconds; default none. Example: `--target-energy-end-s 1900`. Overrides the window end for advisory boundary margin reporting; it is not a production boundary-safety constraint.

3. Path mode

--path-mode

Short form: none. Choices: `geometry-only, phase-aware`; default `geometry-only`. Geometry-only samples a surface great circle; phase-aware

requests TauP geographic ray paths. Example: `--path-mode phase-aware`.

--path-samples

Short form: none. Type: integer; default 121. Applies to geometry-only great-circle sampling. Example: `--path-samples 241`. It does not resample TauP paths and does not establish waveform accuracy.

4. TauP and partial coverage

--taup-model

Short form: none. Type: TauP model name; default `prem`; applies to phase-aware only. Example: `--taup-model prem`. An unavailable model/phase is an error in strict coverage.

--taup-resample

Short form: none. Boolean flag; default false; phase-aware only. Example: `--taup-resample`. Records the choice in each path record; length remains the `taup_raypath_polyline_estimate`.

--ray-param-tol

Short form: none. Type: float; default `1e-06`; phase-aware only. Example: `--ray-param-tol 1e-6`. It is passed to the TauP path request and recorded in metadata.

--cmb-near-depth-tolerance-km

Short form: none. Type: float km; default `25.0`; phase-aware only. Example: `--cmb-near-depth-tolerance-km 20`. Controls the sampled CMB-near proxy, not a physical interface intersection.

--allow-partial-phase-coverage

Short form: none. Boolean flag; default false (strict). Example: `--allow-partial-phase-coverage`. Strict mode fails after listing every missing phase/station pair; partial mode retains only returned requested phases and adds a no-substitution warning/inventory.

5. Target regions

--target-region

Short form: none. Type: strict YAML path; default none. Example: `--target-region target_circle.yaml`. Supports only circle, polygon, or corridor schemas. Unknown/missing fields fail; it changes coverage constraints.

6. Boundary time

--boundary-speed-upper-km-s

Short form: none. Type: positive float km/s; default none. Example: `--boundary-speed-upper-km-s 14`. Produces a surface-arc proxy marked `heuristic_not_conservative`; it is advisory and TauP is forbidden here.

7. Center/gamma search and refinement

--latitude-range

Short form: none. Format MIN,MAX; default `-90,90`. Example: `--latitude-range -20,20`. Defines the center-latitude search interval.

--longitude-range

Short form: none. Format MIN,MAX; default `-180,180`. Example: `--longitude-range 120,180`. Defines the center-longitude interval; polar equivalents are deduplicated deterministically.

--gamma-range

Short form: none. Format MIN,MAX; default `0,360`. Example: `--gamma-range 0,90`. Defines the whole-system gamma search interval.

--coarse-latitude-step

Short form: none. Type: float degrees; default `10.0`. Example: `--coarse-latitude-step 5`. First-pass latitude grid spacing.

--coarse-longitude-step

Short form: none. Type: float degrees; default `10.0`. Example: `--coarse-longitude-step 5`. First-pass longitude grid spacing.

--coarse-gamma-step

Short form: none. Type: float degrees; default `15.0`. Example: `--coarse-gamma-step 10`. First-pass gamma grid spacing.

--local-latitude-step

Short form: none. Type: float degrees; default 2.0. Example: `--local-latitude-step 1`. Local refinement latitude spacing.

--local-longitude-step

Short form: none. Type: float degrees; default 2.0. Example: `--local-longitude-step 1`. Local refinement longitude spacing.

--local-gamma-step

Short form: none. Type: float degrees; default 3.0. Example: `--local-gamma-step 1`. Local refinement gamma spacing.

--final-latitude-step

Short form: none. Type: float degrees; default 0.5. Example: `--final-latitude-step 0.25`. Final latitude spacing around top candidates.

--final-longitude-step

Short form: none. Type: float degrees; default 0.5. Example: `--final-longitude-step 0.25`. Final longitude spacing around top candidates.

--final-gamma-step

Short form: none. Type: float degrees; default 0.5. Example: `--final-gamma-step 0.25`. Final gamma spacing around top candidates.

8. Scoring and coverage

--weights

Short form: none. Type: strict YAML path; default none. Example: `--weights weights.yaml`. Allowed keys are `coverage`, `external_margin`, `endpoint_margin`, `cost`; all must be non-negative and one positive. It replaces score weights, not hard constraints.

--minimum-path-coverage

Short form: none. Type: float in (0,1]; default 1.0. Example: `--minimum-path-coverage 0.9`. Lower values allow partial target/path coverage; values outside the interval fail.

9. NEX, MPI, and cost

--available-ranks

Short form: none. Format: comma-separated total ranks; default uses external `Par_file` NPROC if present, otherwise profile 2,8,12. Example: `--available-ranks 8,12`. Suggestions label only canonical NEX=96 at 2/8/12 as `project_validated`.

--nex

Short form: none. Format `XIXETA`; repeatable; default uses external `Par_file` NEX if present, otherwise profile 96x96. Example: `--nex 96x96 -nex 192x96`. Each value is checked against the selected planning physics branch.

--max-compute-cost

Short form: none. Type: float; default none. Example: `--max-compute-cost 2304`. Filters suggestions by the planner's lateral-work-per-rank proxy; it is not a runtime forecast.

10. Optional Par_file

--par-file

Short form: none. Type: any readable `Par_file` path; default none. Example: `--par-file /work/case/DATA/Par_file`. Reads NEX/NPROC and compatibility flags; it never locates SPECFEM or verifies source/patch status. Omit it to use profile planning defaults marked `builtin_profile`.

11. Output and safety

There is no `--log` option. User-facing errors go to standard error; ordinary planning is quiet except for generated files. Capture shell `stdout/stderr` when an external workflow needs a persistent log.

--output

Short form: none. Type: directory path; required. Example: `--output plan_out`. Must not exist. On success it receives `candidates.json`, `candidates.csv`, `geometry_audit.json`, `boundary_time_audit.json`, `run_manifest.json`, `recommended_Par_file.inc`, `report.md`, `map.png`, and `globe.png`. Geometry-only and phase-aware modes write the same file names; phase-aware adds requested/provided/missing TauP inventory and TauP path metadata to the structured output. Reusing a directory is a deliberate error. Performance acceptance files such as `summary.json`, `performance_matrix.csv`, profiles and persistent logs are not planner output.

License and limits

This package is GPL-3.0-or-later; see [LICENSE](#) and [third-party notices](#). It supports only canonical $90^\circ \times 90^\circ$ geometry, general classification B. Phase-aware Pdiff/Sdiff is validated, sampling stability is indeterminate, and `boundary_time_production_safe=false`. It neither verifies the user patch nor replaces mesher/database/solver, Stacey, topology, or waveform acceptance. For the concise Event 1 runtime result and detailed evidence, see the [project performance record](#).